



 B.SC. FINAL YEAR PROJECT DEFENSE

SafeSurf AI

Intelligent Cybersecurity Threat Detection System

Centralized heuristic threat analysis, external API intelligence integration, and browser-level perimeter defense for web, email, and wireless environments.



Academic Presentation

FACULTY & INSTITUTION

Faculty of Engineering & Technology

Mirpur University of Science and Technology
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Mirpur, Azad Jammu & Kashmir

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Introduction & Problem Statement

Phishing & Domain Impersonation

Phishing domains spoof legitimate institutional services, tricking end-users via visual similarity.

Opaque Link Redirection

Email links use obfuscated URLs and multi-stage redirects to bypass standard spam filters.

Unclear WiFi Security Posture

Users connect to wireless access points without understanding underlying cipher configurations.

THREAT VULNERABILITY FLOW

 End User Interaction



 Suspicious Website / Email Link / Open WiFi



 Exposure to Cyber Threat & Credential Theft

Core Engineering Problem Statement

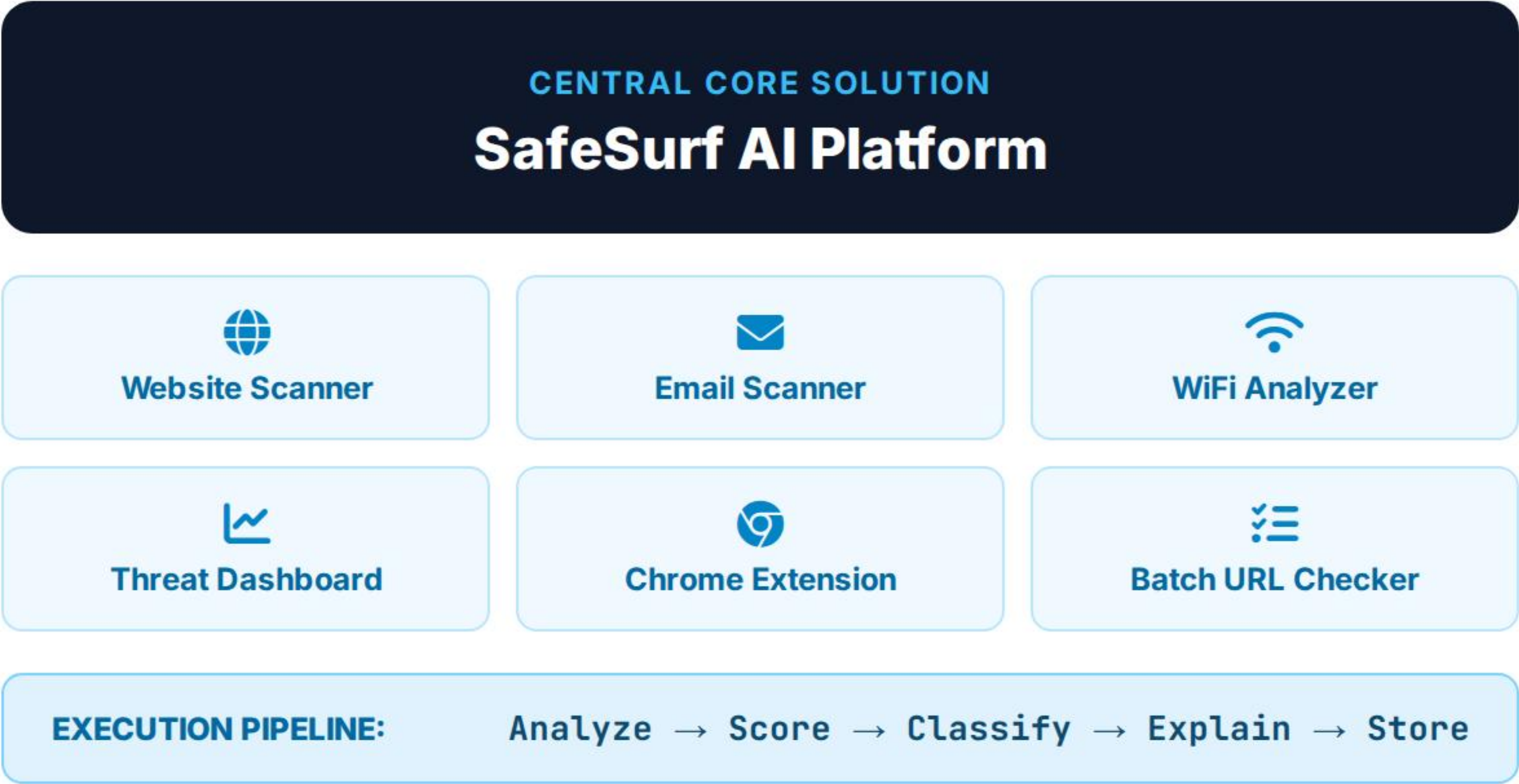
Users require a unified, automated, and centralized preliminary security mechanism to evaluate threat vectors in real time across browser, network, and communication channels.

Objectives & Proposed Solution

System Scope

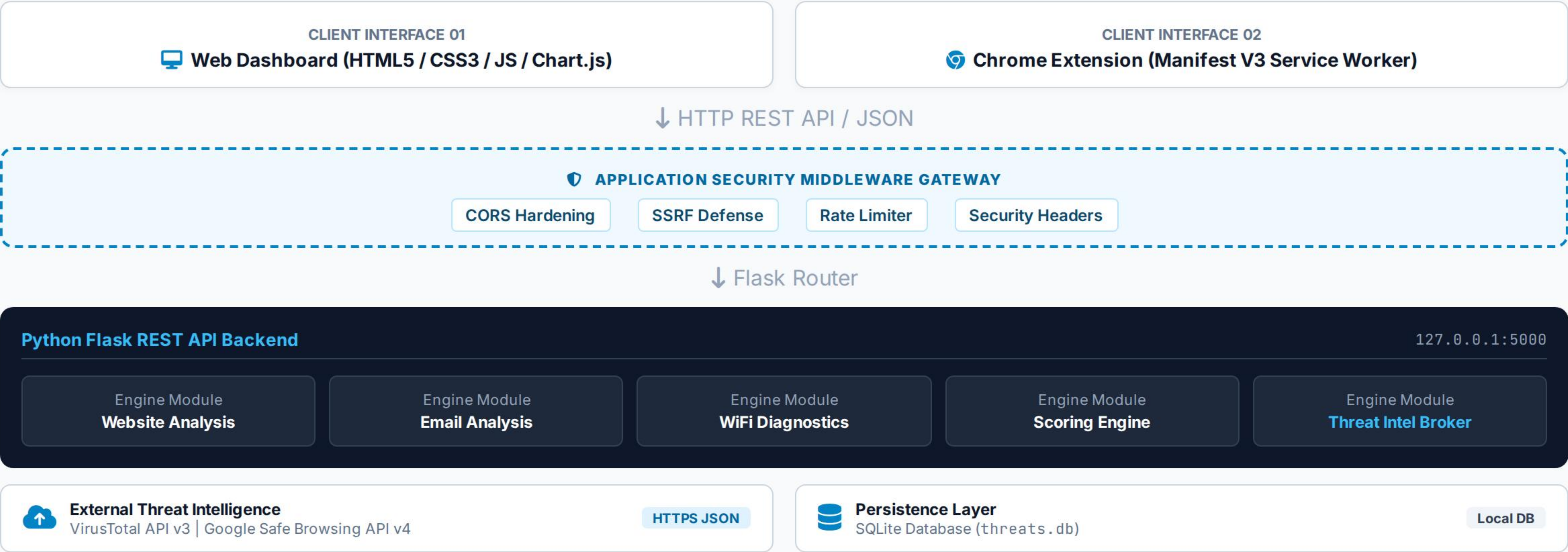
Core Project Objectives

- ✓ **Website Heuristics:** Detect suspicious URL structures, brand impersonation, and missing SSL.
- ✓ **Email Parsing:** Extract embedded links from message bodies for preliminary threat inspection.
- ✓ **WiFi Diagnostics:** Analyze active OS network connection security parameters and cipher protocols.
- ✓ **Threat Intel Aggregation:** Integrate VirusTotal v3 and Google Safe Browsing v4 APIs.
- ✓ **Browser Integration:** Provide real-time page inspection via Chrome Extension (Manifest V3).



End-to-End System Architecture

REST Micro-Architecture



Website Security Analysis & Scoring

Inspection Mechanics

URL Input

Validation

Domain Analysis

Heuristics

Threat Intel

Scoring

Security Result

 Evaluated Security Indicators

 HTTPS / SSL Status

 URL Length & Entropy

 Special Symbols (@, -, %)

 IP-Based Hostnames

 Brand Impersonation

 Reputation Data

OUTPUT VERDICT METRICS

Automated Security Classification

SCORE RANGE

0 – 100

STATUS VERDICT

SAFE / MODERATE / CRITICAL

 Explanatory Feedback

Generates specific heuristic breakdown warnings and recommended remedial actions for end-users.

Email Phishing & WiFi Security Modules

Email Phishing Detection

ANALYTICAL SEQUENCE:

Input → Content Parsing → URL Extraction → Security Analysis → Threat Result

- ✓ Header and sender metadata structure review.
- ✓ Regex parsing engine isolates embedded hyperlinks.
- ✓ Extracted links piped directly to security heuristics pipeline.

 Coverage Purpose

Prevents credential theft originating from suspicious email call-to-action buttons.

WiFi Security Diagnostics

DIAGNOSTIC SEQUENCE:

Connected AP → OS Interface Query → Encryption Check → Security Score

NATIVE OPERATING SYSTEM INTERFACING

Win: netsh wlan show interfaces
Linux: nmcli dev wifi

 Technical Assurance

Queries active OS wireless interface parameters directly. Does not generate synthetic network data.

Browser Security with Chrome Extension

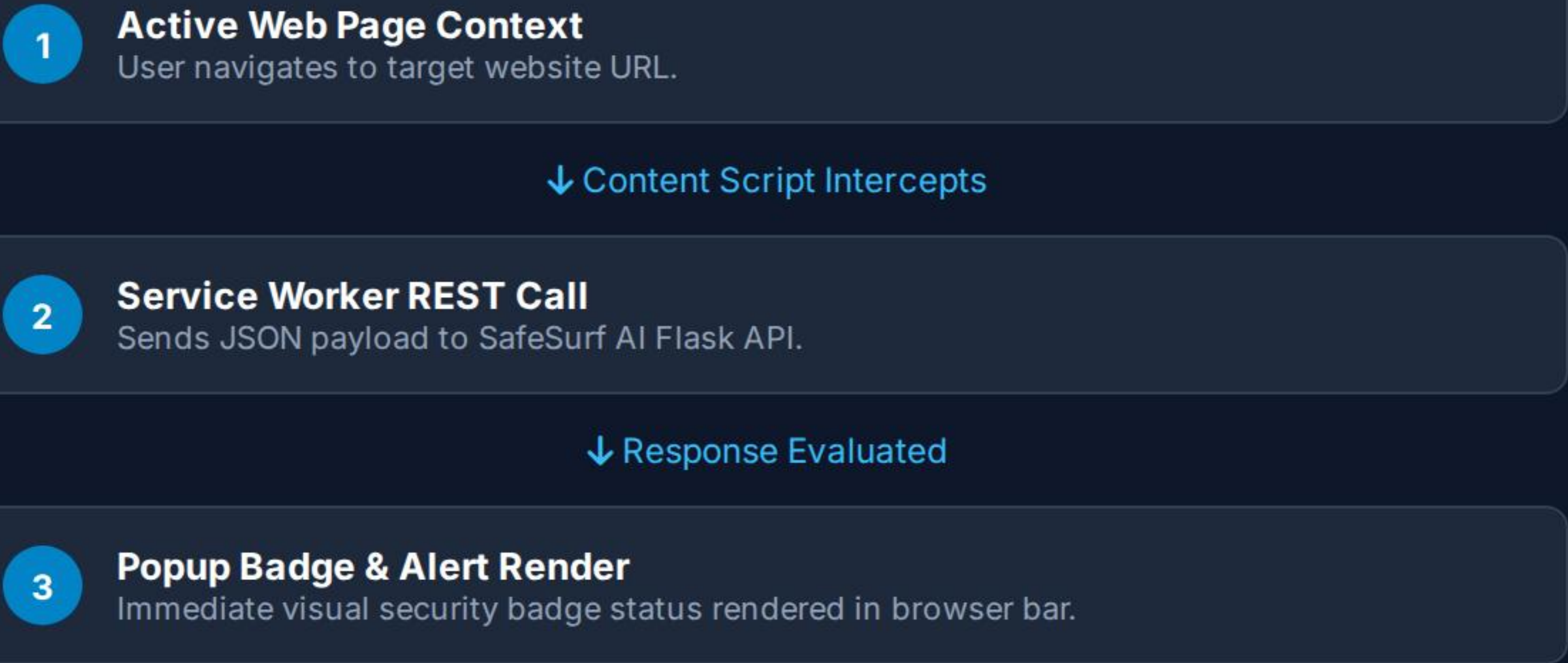
Perimeter Protection

Enables active page inspection directly from the browser window without requiring manual dashboard URL submission.

Architecture Components

Service Worker	Content Script
Popup UI	Local Cache

EXTENSION COMMUNICATION FLOW



Application Security Implementation

OWASP Aligned

OWASP A10

SSRF Protection
Blocks outbound requests targeting loopback (127.0.0.1) and private subnets.

OWASP A01

Restricted CORS
Strict origin white-listing prevents unauthorized cross-site API access.

OWASP A05

Rate Limiting
Restricts automated API abuse and denial-of-service spam attempts.

OWASP A05

Security Headers
Injects HSTS, X-Frame-Options, and CSP headers on API responses.

OWASP A05

Localhost Binding
Flask server bound explicitly to loopback interfaces during testing.

OWASP A03

Path Traversal Defense
Sanitizes incoming file paths and user inputs against directory traversal.

OWASP A07

Environment Secrets
Isolates external VirusTotal / Safe Browsing keys in .env configuration.

OWASP A05

Debug Mode Disabled
Disables interactive debuggers to prevent remote code execution risks.

System Implementation & Database Schema

SQLite Storage

Table: threats

threats.db

Column	Data Type	Description
id	INTEGER PRIMARY KEY	Unique threat record ID
type	TEXT	URL / Email / WiFi
url / target	TEXT	Target domain or vector
score	INTEGER	Calculated score (0-100)
threat_level	TEXT	Low / Medium / High / Critical
details	TEXT / JSON	Heuristic warning breakdown
timestamp	DATETIME	Creation timestamp

BACKEND ENVIRONMENT

Python 3.x + Flask REST Framework

COMMUNICATION PROTOCOL

Asynchronous HTTP / JSON API

FRONTEND RENDERING

HTML5, CSS3, JavaScript ES6, Chart.js

Testing & Functional Evaluation Matrix

Black Box Testing

Module	Test Scenario	Execution Vector	Expected & Observed Result	Status
Website	Valid HTTPS URL	Standard Domain Query	Calculates clean score (SAFE verdict)	PASS
Website	Suspicious Phishing URL	IP / Keyword Spoof URL	High risk flags & threat warnings generated	PASS
Email	Body with embedded links	Raw text payload	Regex extracts links & passes to heuristic engine	PASS
WiFi	Connected AP Query	netsh / nmcli CLI	Extracts active SSID encryption & scores safety	PASS
Extension	Active browser tab check	Service worker alert	Renders threat status badge in browser popup	PASS
Security Layer	SSRF to 127.0.0.1	Loopback URL submission	Middleware blocks request with 403 Forbidden	PASS

Testing Verification Summary

All core modules, database logging, and API security controls verified across functional test cases. External threat API results depend on valid API keys and external service availability.

Project Conclusion

SafeSurf AI successfully delivers a centralized cybersecurity threat analysis platform covering Web, Email, and Wireless access vectors.

- ✓ Rule-based & heuristic domain analysis
- ✓ Threat intelligence API integration
- ✓ Active browser protection via Chrome Extension

Future Enhancements

- **Machine Learning Classifiers:** Train Random Forest / XGBoost models on labeled phishing datasets.
- **Expanded Threat Feeds:** Integrate AlienVault OTX & AbuseIPDB APIs.
- **Cloud Microservices Deployment:** Containerize application using Docker for multi-tenant scalability.

THANK YOU

SafeSurf AI Project Defense — Department of Computer Systems Engineering

Questions & Answers